

Data Science Tasks & Instructions

- **TASK 1: Professional LinkedIn Announcement**
 - **Objective:** Share your official internship acceptance on your professional network.
 - **Requirements:** Post a high-resolution copy of your official **Internship Offer Letter** on LinkedIn. Craft a compelling caption detailing your learning goals, tag **Progree**, and use hashtags: `#datascience` `#python` `#analytics` `#progree`
- **TASK 2: Automated Unstructured Web Scraping & Data Extraction Pipeline**
 - **Objective:** Build an automated system pipeline harvesting data records from remote web environments.
 - **Requirements:** Write an automated data collection script using Python (BeautifulSoup or Scrapy). Pull raw text listings down from an unstructured public source, extract target core parameters using text filtering logic, clean out whitespace artifacts, and write the final output to a database file or a structured CSV.
- **TASK 3: Unsupervised Machine Learning Customer Segmentation (Clustering)**
 - **Objective:** Apply multi-variable clustering models to group population data by behavior.
 - **Requirements:** Ingest transaction or consumer metric datasets into Python using Scikit-Learn. Perform data standardization and execute Principal Component Analysis (PCA) to reduce data dimensions. Optimize cluster counts using the Elbow Method and fit a K-Means clustering model, visualizing distinct customer segment distributions.
- **TASK 4: Mini Project - Time-Series Forecasting Model & Advanced Model Interpretability Dashboard**
 - **Objective:** Fit mathematical time-series models to historic metrics and map variable impacts using interpretability layers.
 - **Requirements:** Load a historical sequential business or climate dataset. Check for stationarity trends, apply seasonal differencing pipelines, and train a forecasting model (SARIMA or Prophet) to predict future values with confidence intervals. Implement a model interpretability wrapper (SHAP or LIME) to measure

exact feature importance weights, exporting findings into a comprehensive data science whitepaper (5–6 pages).

PROGRESS